Two orbit columns under the lazy member index

2026-10-18. Study `research/ecbench_lazy_columns2_20261018/`, protocol `PROTOCOL.md` (frozen in commit `760c672a5` before any claim data). Class: **engineering**. Stacked on PR #2158 (oracle `mitm-frobenius-lazy-counted`). Sessions ran on commit `760c672a5` (`git_dirty: false`), `ecbench` binary `ae108d54…`, 4-vCPU Xeon VM, isolation L0 (counts stand; wall times descriptive only).

Result

Question: now that the lazy oracle has removed the member-index set-up that grew with the number of columns, is `columns = 2` cheaper than the per-curve column count frozen on 2026-10-14 under the eager oracle?

Measured (counted `ecbench.gae`, every counter priced, paired by target, 95 % cluster-bootstrap): on one curve of five. Two columns is cheaper at $n = 17$, $a = 1$ (frozen $c = 1$): 1.176 [1.063, 1.297]. It is dearer on the other four: $n = 13$ (frozen $c = 1$) 0.637 [0.630, 0.645]; $n = 23$ ($c = 3$) 0.778 [0.695, 0.874]; $n = 29$, $a = 1$ ($c = 1$) 0.644 [0.625, 0.665]; $n = 37$ ($c = 8$) 0.246 [0.200, 0.309]. Ratios are S(frozen c) ÷ $S(c = 2)$, above 1 meaning two columns is cheaper. The frozen column counts, chosen under the eager oracle, stay the better choice on four of the five curves under the lazy oracle.

All 2 880 records verified, none a lower bound, every audit passed with all replays reproduced (`audits/`), and `rho-frob-aa ÷ rho-frob` is 1 [1, 1] on every curve.

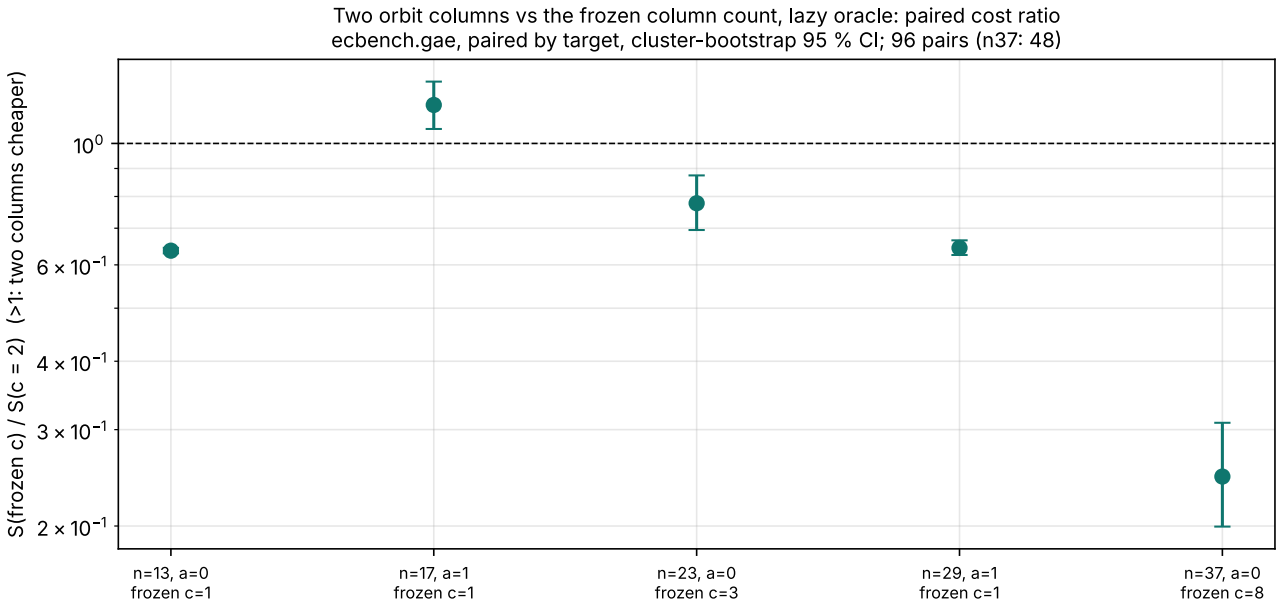


Figure 1: Frozen-c ÷ two-column paired cost ratio per curve. Sessions ECBS1h5f7f47653898 (n13), ECBS1had6c4dfa75fc (n17), ECBS1h88a016cfa7c0 (n23), ECBS1h486ae2ee8d06 (n29), ECBS1hbef0622fd5f3 (n37).

curve (a, n)	frozen c	S, frozen c	S, $c = 2$	frozen ÷ $c2$ (H1)	$c2$ ÷ rho-strong	frozen ÷ rho-strong
0, 13	1	4.667	7.328	0.637 [0.630, 0.645]	0.510 [0.503, 0.516]	0.325 [0.319, 0.330]
1, 17	1	2.483	2.112	1.176 [1.063, 1.297]	0.512 [0.490, 0.535]	0.602 [0.545, 0.665]
0, 23	3	1.265	1.627	0.778 [0.695, 0.874]	1.403 [1.236, 1.578]	1.091 [1.009, 1.176]
1, 29	1	1.830	2.840	0.644 [0.625, 0.665]	0.615 [0.602, 0.627]	0.396 [0.381, 0.411]
0, 37	8	1.044	4.239	0.246 [0.200, 0.309]	15.42 [12.28, 18.61]	3.798 [3.266, 4.403]

Table 1: Mean S (`ecbench table`) and paired ratios (`ecbench compare`, `sessions/*/comparisons/`). 96 pairs per curve (32 targets × 3 rounds); n37: 48. Columns 6 and 7 are cost ÷ the matched strong rho, below 1 meaning the IC arm is cheaper.

What changes (H3)

Two columns need fewer trials than one column (n13, n17, n29) and more than three or eight (n23, n37). Mean trials per solve, frozen c against $c = 2$ (derived: mean of the records' `targets_tried`): n13 12.8 vs 5.8; n17 280.5 vs 89.7; n29 53.5 vs 20.5; n23 802.3 vs 1 415; n37 9 565 vs 53 115. Mean relations (= rank) per solve: 2.0 for $c = 1$, 3.6 for $c = 3$, 7.3 for $c = 8$, 2.9 for $c = 2$. Fewer trials pay for the wider base at n17 only; at n13 and n29 the extra relations cost more than the trials saved.

Distance to rho (reported, not a gate)

`ic-lazy-c2 ÷ rho-strong` is below 1 at n13, n17 and n29 and above 1 at n23 (1.403) and n37 (15.4). The frozen choice is below 1 at n13, n17 and n29 and above 1 at n23 (1.091 [1.009, 1.176]) and n37 (3.80). No ranking against rho is claimed.

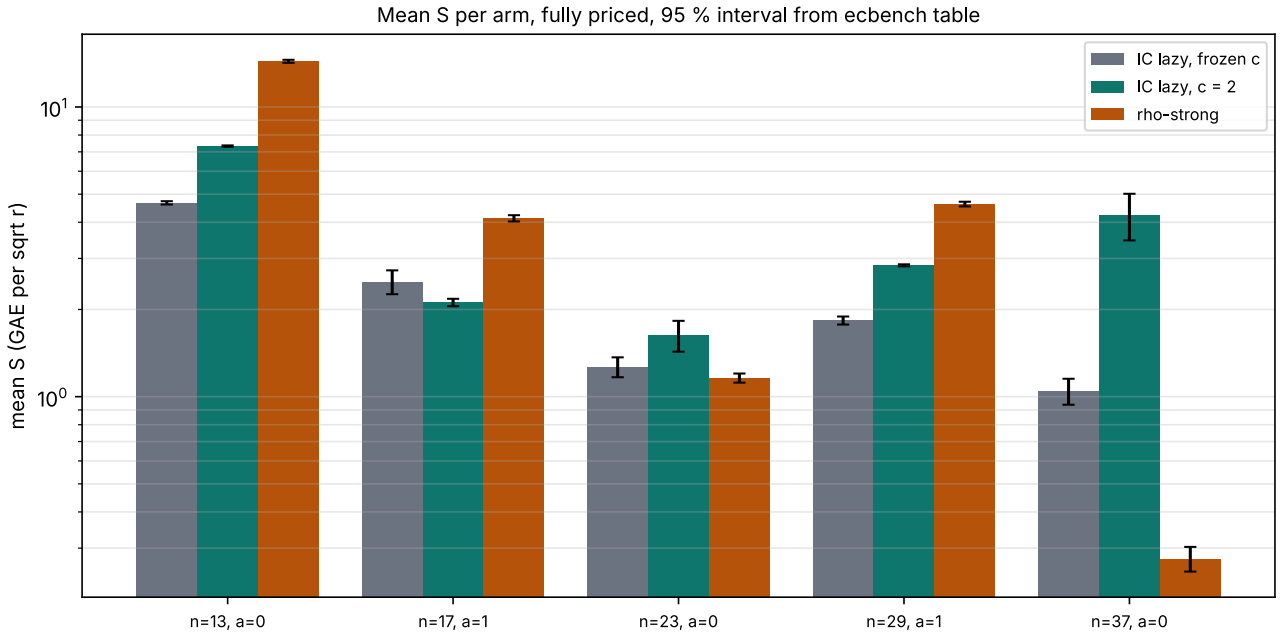


Figure 2: Mean S per arm, 95 % interval from `ecbench table`.

Classification and limits

Engineering. One improvement, on one curve ($n = 17$, $a = 1$, frozen $c = 1$ to $c = 2$, 1.18×), among five pre-declared comparisons; four regress. Choosing $c = 2$ at $n = 17$ after this run is a tuning result on these seeds and needs fresh seeds before it is a rule (protocol §6). The exploratory sweep that suggested two columns (sessions not kept) compared $c = 2$ with $c = 3, 4, 5, 6$ and did not try $c = 1$; the claim shows the frozen $c = 1$ beats $c = 2$ at $n = 13$ and $n = 29$. No exponent, ECC2K-130 or deployed-curve statement follows.

Evidence: `sessions/` (5 ecbench sessions, loose), `audits/`, `tables/`, `logs/`, `figures/plot_figures.py` (lays out recorded values only).